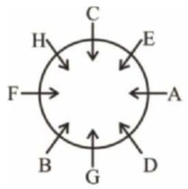
**40. Eight students A, B, C, D, E, F, G and H sit around a circular table, equidistant from each other, facing the center of the table, not necessarily in the same order. B and D sit neither adjacent to C nor opposite to C. A sits in between E and D, and F sits in between B and H. Which one of the following is definitely correct?**

(a) B sits in between A and G
(b) C sits opposite to G
(c) E sits opposite to F
(d) None of the above

**Ans. (d) :**




Hence, 'C' is sitting opposite to 'G'.

**Directions for the following 3 (three) items:**

**Read the following two passages and answer the items that follow the passages. Your answers to these items should be based on the passages only.**

**Passage-1**

**For two or three generations past, ever-increasing numbers of individuals have been living as workers merely, no as human beings. An excessive amount of labour is rule today in every circle of society, with the result that man's spiritual elements cannot thrive. He finds it very difficult to spend his little leisure in serious activities. He does not want to think; or he cannot even if he wants to. He seeks not self-improvement, but entertainment which would enable him to be mentally idle and to forget his usual activities. Therefore, the so-called culture of our age is dependant more on cinema than on theater, more on newspapers, magazines and crime stories than on serious literature.**

**41. The passage is based on the idea that**

(a) Man should not work hard
(b) The great evil of our age is overstrain
(c) Man cannot think well
(d) Man cannot care for his spiritual welfare

**Ans. (b) :** The passage is based on the idea that the great evil of our age is over strain.

**42. Man does not seek self-improvement because he**

(a) is not intellectually capable
(b) has no time to do so
(c) is distracted by materialism
(d) loves amusement and is mentally idle

**Ans. (b) :** According to the passage, man does not seek self improvement because he finds it difficult to engage himself in some serious activity in the little leisure he gets after doing a lot of hard work. So he has no time for self improvement.

**Passage-2**

**The demographic dividend, which has begun in India and is expected to last another few decades, is a great window of opportunity. The demographic dividend is basically a swelling in the working age population, which conversely means that the relative ratio of very young and very old will, for a while, be on the decline From the experience of Irelnad and China, we know that this can be a source of energy and an engine of economic growth. The demographic dividend tends to raise a nation's savings rate since in any nation, it is the working age population that is the main saver. And since the savings rate is an important driver of growth, this should help elevate our growth rate. However, the benefits of demographic dividend depend on the quality of the working age population. And this implies bringing back the importance of education, acquisition of skills and human capital.**

**43. Which of the following would invariably happen in a country, when the demographic dividend has begun to operate?**

(a) The number of illiterate- people will decrease.
(b) The ratio of very old and very young will decrease for a while.
(c) Population growth rate will quickly stabilize.

**Select the correct answer using the code given below.**

(a) 1 and 2 only (b) 2 only
(c) 1 and 3 only (d) 1, 2 and 3

**Ans. (b) :** The ratio of very old and very young will decrease for a while. This would invariably happen in a country, when the demographic dividend begins to operate.

**44. With reference to the passage, which of the following inferences can be drawn?**

1. **Demographic dividend is an essential condition for a country to rapidly increase its economic growth rate.**
2. **Promotion of higher education is an essential condition for a country for its rapid economic growth**

**Select the correct answer using the code given below.**

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

**Ans. (d) :** It is clear from the passage that we know from the experience of Ireland and China that the demographic dividend can be source of energy and an engine of economic growth, but this is an essential condition, which is not mentioned anywhere in the passage, so statement 1 is false.
According to the passage, education is not only essential condition, which leads to economic growth. Along with this, the importance of skill acquisition and human capital is also essential. Therefore, statement (2) is false.

**45. Five friends P, Q, X, Y and Z purchased some notebooks. The relevant information is given below:**

1. **Z purchased 8 notebooks more than X did.**
2. **P and Q together purchased 21 notebooks.**
3. **Q purchased 5 notebooks less than P did.**
4. **X and Y together purchased 28 notebooks.**
5. **P purchased 5 notebooks more than X did.**

**If each notebook is priced ₹ 40, then what is the total cost of all the notebooks?**

(a) ₹ 2,600 (b) ₹ 2,400
(c) ₹ 2,360 (d) ₹ 2, 320

**Ans. (a) :** According to the question,

$Z - X = 8$ ---------(i)
$P + Q = 21$ -------- (ii)
$P - Q = 5$ --------(iii)
$X + Y = 28$ ---------(iv)
$P - X = 5$ ---------(v)

From equation (ii) and (iii)

$2P = 26$
$P = 13, Q = 8$
$X = 8, Y = 20, Z = 16$

Total cost price of notebooks
$= (13 + 8 + 8 + 20 + 16) \times 40$
$= 65 \times 40$
$= ₹\ 2600$

**46. A man started from home at 14:30 hours and drove to village, arriving there when the village clock indicated 15:15 hours. After staying for 25 minutes, he drove back by a different route of length 1.25 times the first route at a rate twice as fast reaching home at 16:00 hours As compared to the clock at home, the village clock is**

(a) 10 minutes slow
(b) 5 minutes slow
(c) 10 minutes fast
(d) 5 minutes fast

**Ans. (d) :** Let the distance of the village from the home be x km.
And speed of the man is V km/hour. It takes a total of 45 minutes for a man to reach the village clock.
According to the question,

$$\frac{x}{V}+\frac{1.25x}{2V}=\frac{65}{60}$$

$3.25x/2V = 65/60$

$$\frac{x}{V}=\frac{65}{3.25\times30}\text{ hours}$$

$$t=\frac{65\times60}{3.25\times30}$$

$t = 40$ minute

Hence, the clock of the village is 5 minute faster.

**47. A person X wants to distribute some pens among six children A, B, C, D, E and F. Suppose A gets twice the number of pens received by B, three times that of C, four times that of D, five times that of E and six times that of F. What is the minimum number of pens X should buy so that the number of pens each one gets is an even number?**

(a) 147 (b) 150
(c) 294 (d) 300

**Ans. (c) :** According to the question,

$A = 2B = 3C = 4D = 5E = 6F$

$$\frac{A}{60}=\frac{B}{30}=\frac{C}{20}=\frac{D}{15}=\frac{E}{12}=\frac{F}{10}$$

$\because$ Total sum $= (60 + 30 + 20 + 15 + 12 + 10)\ x = 147x$
If $x = 1$
Total number of pens = 147
But total number of D = 15 (Odd number)
So we take $x = 2$
Total number of pen $= 147 \times 2 = 194$

**48. Six persons A, B, C, D, E and F are sitting equidistant from each other around a circular table (facing the centre of the table).**

**Consider the Question and two Statements given below:**

**Question: Who is sitting on the immediate left of A?**

**Statement-1: B is sitting opposite to C and D is sitting opposite to E.**

**Statement-2: F is sitting on the immediate left of B.**

**Which one of the following is correct in respect of the Question and the Statements?**

(a) Statement-1 alone is sufficient to answer the Question
(b) Statement-2 alone is sufficient to answer the Question